

CNN FIRE DETECTION

Fire detection is an important application of Artificial Intelligence (AI) that helps protect lives and property by identifying fire at an early stage. In this project, a Convolutional Neural Network (CNN) is used to detect fire from images. The model is trained using a dataset of fire and non-fire images, enabling it to accurately classify whether an input image contains fire.

The project is implemented using Python with deep learning libraries such as TensorFlow, Keras, and OpenCV. The CNN model automatically learns important features from the training images and provides reliable predictions. This system offers a fast, simple, and effective solution for early fire detection and can be applied in industries, forests, public places, and smart surveillance systems to improve safety and reduce fire-related risks.